Understanding Analysis, 2nd Edition
Chapter 6 - Section 6.4 - Series of Functions

1. Supply the details for the proof of the Weierstrass M-Test (Corollary 6.4.5).

Solution: We have for every fixed x

$$0 \leq |f_n(x)| \leq M_n$$

and $\sum_{n=1}^{\infty} M_n$ converges. This satisfies the conditions for the Comparison Test, so $\sum_{n=1}^{\infty} |f_n(x)|$ converges.

Then by the Absolute Convergence Test, $\sum_{n=1}^{\infty} |f_n(x)|$ converges implies $\sum_{n=1}^{\infty} f_n(x)$ converges.

It remains to prove that the convergence is uniform. $\sum_{n=1}^{\infty} M_n$ converges so it satisfies the Cauchy Criterion for Series (Theorem 2.7.2): for any $\epsilon > 0$, there exists N such that for all $n > m \geq N$:

$$M_{m+1} + M_{m+2} + ... + M_n < \epsilon$$

Reusing the same N, we can prove that $\sum_{n=1}^{\infty} f_n(x)$ satisfies the Cauchy Criterion for Uniform Convergence of Series (Theorem 6.4.4). Because $|f_n(x)| \leq M_n$, then

$$\begin{aligned} |f_{m+1}(x) + f_{m+2}(x) + ... + f_n(x)| &\leq |f_{m+1}(x)| + |f_{m+2}(x)| + ... + |f_n(x)| \\ &\leq M_{m+1} + M_{m+2} + ... + M_n \\ &< \epsilon \end{aligned}$$

Since the same N works for any x (because the bound M_n is independent of x), the convergence is uniform.

2. Short proof or counterexample.

   (a) If $\sum_{n=1}^{\infty} g_n$ converges uniformly, then (g_n) converges uniformly to zero.

   (b) If $0 \leq f_n(x) \leq g_n(x)$ and $\sum_{n=1}^{\infty} g_n$ converges uniformly, then $\sum_{n=1}^{\infty} f_n$ converges uniformly.

   (c) If $\sum_{n=1}^{\infty} f_n$ converges uniformly on A, then there exist constants M_n such that $|f_n(x)| \leq M_n$ for all $x \in A$ and $\sum_{n=1}^{\infty} M_n$ converges.

Solution:

(a) True. This is analogous with the nth-term test for series of numbers (Theorem 2.7.3).

$\sum_{n=1}^{\infty} g_n$ converges uniformly so there exists N such that for all $n > m > N$ and for all x

$$|g_{m+1}(x) + ... + g_n(x)| < \epsilon$$

The same N works to prove (g_n) converges uniformly. Set $n = m + 1$.

Alternatively, without using the Cauchy Criterion, let $s_k = \sum_{n=1}^{k} g_n(x)$. Then

$$\lim_{k\to\infty} s_k - s_{k-1} = \lim_{k\to\infty} s_k - \lim_{k\to\infty} s_{k-1} = 0$$

And note that $s_k - s_{k-1} = g_k$, so $\lim_{k\to\infty} s_k - s_{k-1} = \lim_{k\to\infty} g_k(x) = 0$.

The important intuition here is that uniform convergence allows us to reuse many results from series of numbers, because any bound works uniformly so we may treat $g_n(x)$ for all x like a single entity.

(b) True. This is analogous to the Comparison Test for series of numbers (Theorem 2.7.4).

Like part (a), but we can get rid of the absolute values because it's all non-negative

$$g_{m+1}(x) + ... + g_n(x) < \epsilon$$

then reusing the same N and the problem's assumption that $0 \leq f_n(x) \leq g_n(x)$

$$f_{m+1}(x) + ... + f_n(x) \leq g_{m+1}(x) + ... + g_n(x) < \epsilon$$

(c) False. This is analogous to conditional convergence for series of numbers (Theorem 2.7.8).

Take $f_n(x) = \frac{(-1)^{n+1}}{n}$ for all x. Then $\sum_{n=1}^{\infty} f_n$ is the alternating harmonic series for all x so it converges uniformly.

But $M_n \geq \frac{1}{n}$ so $\sum_{n=1}^{\infty} M_n$ diverges by comparison test with $\sum_{n=1}^{\infty} \frac{1}{n}$.

3. (a) Show that

$$g(x) = \sum_{n=0}^{\infty} \frac{\cos(2^n x)}{2^n}$$

is continuous on all of $\mathbf{R}$.

(b) The function g was cited in Section 5.4 as an example of a continuous nowhere differentiable function. What happens if we try to use Theorem 6.4.3 to explore whether g is differentiable?

Solution:

(a) We have $|\cos(2^n x)| \leq 1$ for any n and x. Then

$$\left|\frac{\cos(2^n x)}{2^n}\right| \leq \frac{1}{2^n}$$

for all x and $\sum_{n=0}^{\infty} \frac{1}{2^n} = 2$. This satisfies the Weierstrass M-test, so $\sum_{n=0}^{\infty} \frac{\cos(2^n x)}{2^n}$ converges uniformly.

By Theorem 6.4.2 (Term-by-term Continuity Theorem), since $\frac{\cos(2^n x)}{2^n}$ is continuous for any n and $\sum_{n=0}^{\infty} \frac{\cos(2^n x)}{2^n}$ converges uniformly, then g is continuous.

(b) The series of derivative is

$$\sum_{n=0}^{\infty} -\sin(2^n x)$$

This fails to converge uniformly so we can't apply Theorem 6.4.3. For example, take $x = \pi/3$, then the terms are

$$-\sin\left(\frac{2^n \pi}{3}\right) = \begin{cases} -\frac{\sqrt{3}}{2}, & n = 0 \\ (-1)^n \frac{\sqrt{3}}{2}, & n \geq 1 \end{cases}$$

Since the terms fail to approach 0, the series fails to converge.

This does not prove that g is not differentiable; it only means we can't use Theorem 6.4.3 to conclude differentiability.

4. Define

$$g(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(1+x^{2n})}$$

Find the values of x where the series converge and show that we get a continuous function on this set.

Solution: We have

$$\frac{x^{2n}}{(1+x^{2n})} = \frac{x^{2n}+1-1}{(1+x^{2n})} = 1 - \frac{1}{1+x^{2n}}$$

For $x \geq 1$ or $x \leq -1$, the terms converge to 1, so the series diverges.

For $x \in (-1,1)$ the series converges but not uniformly so we can't directly apply Theorem 6.4.2. But we can use the same trick as problem 5(b). Choose r such that $|x| < r < 1$, then on $[-r,r]$

$$0 \leq \frac{x^{2n}}{1+x^{2n}} \leq x^{2n} \leq r^{2n}$$

Since $\sum_{n=0}^{\infty} r^{2n}$ is a convergent geometric series, we may apply the Weierstrass M-Test to conclude that $\sum_{n=0}^{\infty} \frac{x^{2n}}{(1+x^{2n})}$ converges uniformly on $[-r,r]$.

Since each term of the series is continuous, we may apply the Term-by-term Continuity Theorem to conclude that g is continuous on $[-r,r]$ hence continuous on x. Since this argument works for any $x \in (-1,1)$, g is continuous on $(-1,1)$.

5. (a) Prove that

$$h(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2} = x + \frac{x^2}{4} + \frac{x^3}{9} + ...$$

is continuous on $[-1,1]$.

(b) The series

$$f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n} = x + \frac{x^2}{2} + \frac{x^3}{3} + ...$$

converges for every x in the half-open interval $[-1,1)$ but does not converge when $x = 1$. For a fixed $x_0 \in (-1,1)$, explain how we can still use the Weierstrass M-Test to prove that x is continuous at x_0.

Solution:

(a) We have for $x \in [-1, 1]$

$$\left|\frac{x^n}{n^2}\right| \leq \frac{1}{n^2}$$

and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. This satisfies the conditions for the Weierstrass M-Test, hence the series converges uniformly.

We have $\frac{x^n}{n^2}$ continuous for every n. Combined with the uniform convergence above, we may apply the Term-by-term Continuity Theorem to conclude that h is continuous.

(b) For any $x_0 \in (-1, 1)$, we can find an r such that $|x_0| < r < 1$. Now we can bound the function with r. For every $x \in [-r, r]$:

$$\frac{x^n}{n} \leq \frac{|x^n|}{n} \leq r^n$$

Since $|r| < 1$, the geometric series $\sum_{n=1}^{\infty} r^n$ converges and we can apply the Weierstrass M-Test to conclude that $\sum_{n=1}^{\infty} \frac{x^n}{n}$ converges uniformly on $[-r, r]$.

Since each term of the series is continuous, we may apply the Term-by-term Continuity Theorem to conclude that $f(x)$ is continuous on $[-r, r]$. Since $x_0 \in [-r, r]$, f is continuous on x_0 as well.

6. Let

$$f(x) = \frac{1}{x} - \frac{1}{x+1} + \frac{1}{x+2} - \frac{1}{x+3} + \frac{1}{x+4} - ...$$

Show f is defined for all $x > 0$. Is f continuous on $(0, \infty)$? How about differentiable?

Solution: By the Alternating Series Test, the series converges for all $x > 0$.

It's continuous on $(0, \infty)$. We use the same trick as problem 4 and 5. For any $x \in (0, \infty)$, there exists a, b such that $0 < a \leq x \leq b$. We'll prove that f converges uniformly on $[a, b]$. Then by the Term-by-term Continuity Theorem, f is continuous on $[a, b]$.

We'll prove uniform convergence by using the Cauchy Criterion (this is very similar to Exercise 2.7.1). First, recall the following fact about partial sums of an Alternating Series

$$|s_j - s_i| = |\sum_{k=i+1}^{j} (-1)^{k+1} a_k| \leq a_{i+1}$$

Pick N large enough such that $\frac{1}{a+N} < \epsilon$. Note that because the domain is $[a, b]$,

then for any $x \in [a, b]$

$$\frac{1}{x+N} \leq \frac{1}{a+N} < \epsilon$$

Now for any $n > m > N$ and any $x \in [a, b]$, we have

$$|s_n(x) - s_m(x)| \leq \frac{1}{x+N} \leq \frac{1}{a+N} < \epsilon$$

satisfying the Cauchy Criterion.

It's differentiable. We'll show that it satisfies the conditions of the Term-by-term Differentiability Theorem.

Condition #1: We have $f_n(x) = \frac{(-1)^n}{x+n}$ is differentiable with $f_n'(x) = -\frac{(-1)^{n+1}}{(x+n)^2}$.

Condition #2: $\sum_{n=1}^{\infty} f_n'(x)$ converges uniformly by the Weierstrass M-test. The bound is

$$\left|\frac{(-1)^{n+1}}{(x+n)^2}\right| \leq \frac{1}{n^2}$$

and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

Condition #3: We've already established that f is defined for all $x > 0$ (the Theorem only requires a single point $x_0 \in [a, b]$ where f is defined). Now all conditions are satisfied.